

```
In [1]: import pandas as pd
import numpy as np
from sklearn import preprocessing
```

```
In [10]: df=pd.read_csv(r"C:\Users\Admin\Desktop\Engineering\DSBDA practical\archive\iris.csv")
```

```
In [11]: df
```

```
Out[11]:
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

150 rows × 5 columns

```
In [12]: df.head()
```

```
Out[12]:
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

```
In [13]: df.tail()
```

```
Out[13]:
```

	sepal_length	sepal_width	petal_length	petal_width	species
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

```
In [14]: df.index()
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[14], line 1
----> 1 df.index()

TypeError: 'RangeIndex' object is not callable
```

```
In [15]: df.index
```

```
Out[15]: RangeIndex(start=0, stop=150, step=1)
```

```
In [16]: df.columns
```

```
Out[16]: Index(['sepal_length', 'sepal_width', 'petal_length', 'petal_width',
               'species'],
              dtype='object')
```

```
In [17]: df.shape
```

```
Out[17]: (150, 5)
```

```
In [18]: df.columns.values
```

```
Out[18]: array(['sepal_length', 'sepal_width', 'petal_length', 'petal_width',
               'species'], dtype=object)
```

```
In [20]: df.describe(include='all')
```

Out[20]:

	sepal_length	sepal_width	petal_length	petal_width	species
count	150.000000	150.000000	150.000000	150.000000	150
unique	NaN	NaN	NaN	NaN	3
top	NaN	NaN	NaN	NaN	setosa
freq	NaN	NaN	NaN	NaN	50
mean	5.843333	3.054000	3.758667	1.198667	NaN
std	0.828066	0.433594	1.764420	0.763161	NaN
min	4.300000	2.000000	1.000000	0.100000	NaN
25%	5.100000	2.800000	1.600000	0.300000	NaN
50%	5.800000	3.000000	4.350000	1.300000	NaN
75%	6.400000	3.300000	5.100000	1.800000	NaN
max	7.900000	4.400000	6.900000	2.500000	NaN

In [22]: `df['sepal_length']`

Out[22]:

0	5.1
1	4.9
2	4.7
3	4.6
4	5.0
...	
145	6.7
146	6.3
147	6.5
148	6.2
149	5.9

Name: sepal_length, Length: 150, dtype: float64

In [23]: `df.sort_index(axis=1,ascending=False)`

Out[23]:

	species	sepal_width	sepal_length	petal_width	petal_length
0	setosa	3.5	5.1	0.2	1.4
1	setosa	3.0	4.9	0.2	1.4
2	setosa	3.2	4.7	0.2	1.3
3	setosa	3.1	4.6	0.2	1.5
4	setosa	3.6	5.0	0.2	1.4
...	...	...	...	...	...
145	virginica	3.0	6.7	2.3	5.2
146	virginica	2.5	6.3	1.9	5.0
147	virginica	3.0	6.5	2.0	5.2
148	virginica	3.4	6.2	2.3	5.4
149	virginica	3.0	5.9	1.8	5.1

150 rows × 5 columns

In [24]: `df.sort_values(by="sepal_length")`

Out[24]:

	sepal_length	sepal_width	petal_length	petal_width	species
13	4.3	3.0	1.1	0.1	setosa
42	4.4	3.2	1.3	0.2	setosa
38	4.4	3.0	1.3	0.2	setosa
8	4.4	2.9	1.4	0.2	setosa
41	4.5	2.3	1.3	0.3	setosa
...	...	...	...	...	...
122	7.7	2.8	6.7	2.0	virginica
118	7.7	2.6	6.9	2.3	virginica
117	7.7	3.8	6.7	2.2	virginica
135	7.7	3.0	6.1	2.3	virginica
131	7.9	3.8	6.4	2.0	virginica

150 rows × 5 columns

In [25]: `df.iloc`Out[25]: `<pandas.core.indexing._iLocIndexer at 0x1ed5a1f54a0>`

```
In [28]: df[0:3]
```

```
Out[28]:
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa

```
In [32]: df.loc[:, ["sepal_length", "sepal_width"]]
```

```
Out[32]:
```

	sepal_length	sepal_width
0	5.1	3.5
1	4.9	3.0
2	4.7	3.2
3	4.6	3.1
4	5.0	3.6
...	...	...
145	6.7	3.0
146	6.3	2.5
147	6.5	3.0
148	6.2	3.4
149	5.9	3.0

150 rows × 2 columns

```
In [33]: df.iloc[:, :4]
```

Out[33]:

	sepal_length	sepal_width	petal_length	petal_width
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2
...	...	...	...	...
145	6.7	3.0	5.2	2.3
146	6.3	2.5	5.0	1.9
147	6.5	3.0	5.2	2.0
148	6.2	3.4	5.4	2.3
149	5.9	3.0	5.1	1.8

150 rows × 4 columns

In [34]: `df.iloc[3:5,0:2]`

Out[34]:

	sepal_length	sepal_width
3	4.6	3.1
4	5.0	3.6

In [35]: `df.isnull()`

Out[35]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
145	False	False	False	False	False
146	False	False	False	False	False
147	False	False	False	False	False
148	False	False	False	False	False
149	False	False	False	False	False

150 rows × 5 columns

In [36]: df.notnull

Out[36]: <bound method DataFrame.notnull of

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
..	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

[150 rows x 5 columns]>

In [37]: df=pd.read_csv(r"C:\Users\Admin\Desktop\Engineering\DSBDA practical\archive\iris.csv")

In [38]: df.head

```
Out[38]: <bound method NDFrame.head of
width    species
0         5.1      3.5      1.4      0.2    setosa
1         4.9      3.0      1.4      0.2    setosa
2         4.7      3.2      1.3      0.2    setosa
3         4.6      3.1      1.5      0.2    setosa
4         5.0      3.6      1.4      0.2    setosa
..        ...      ...      ...      ...      ...
145        6.7      3.0      5.2      2.3  virginica
146        6.3      2.5      5.0      1.9  virginica
147        6.5      3.0      5.2      2.0  virginica
148        6.2      3.4      5.4      2.3  virginica
149        5.9      3.0      5.1      1.8  virginica

[150 rows x 5 columns]>
```

In []: